The Wrong Kind of Convergence: School-Based Tutoring, Private Tutoring, and Rural–Urban Inequality in China's Double Reduction Era

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Abstract

Private supplementary tutoring—shadow education—has expanded across East Asia and other middle-income countries, and access is uneven: urban and affluent families buy more tutoring than rural and poor families. China's 2021 "double reduction" policy restricted private academic tutoring while expanding school-based after-school services, creating a critical case for asking whether public provision can substitute for private tutoring and reduce inequality. Using the China Family Panel Studies child module for 2018, 2020, and 2022, this paper examines the rural–urban tutoring gap among children aged 10–15. The aggregate gap in any tutoring fell from 19.8 to 6.5 percentage points. Decomposition shows that the school-based gap fell from 14.0 to 2.2 points, while the private gap fell only from 7.5 to 4.9 points; conditional on family background, the private gap changed little and the change is not statistically significant. Rural households also continued to spend far less on tutoring. The aggregate convergence therefore reflects public school-based provision—much of which predates 2021—rather than convergence in the private market. Public provision can narrow access gaps without reducing market-based inequality, with implications for regulating shadow education in developing countries.

Keywords: shadow education; private tutoring; school-based tutoring; rural–urban inequality; double reduction; China; education policy

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1. Introduction

Private supplementary tutoring—shadow education—has expanded rapidly across East Asia and other middle-income countries. Families purchase additional instruction to prepare for high-stakes examinations, to compensate for perceived gaps in school quality, and to keep pace with peers. Because tutoring is bought in a market, access is uneven. Urban and affluent households can afford more hours, better providers, and more expensive forms of instruction, while rural and poor households face higher travel costs, thinner markets, and tighter budgets. The result is a form of educational inequality that operates alongside, and often reinforces, inequality in formal schooling.

China is a critical case for understanding this relationship. In 2021 the government launched the "double reduction" policy, which restricted for-profit academic tutoring in compulsory education, prohibited academic tutoring on weekends and holidays, limited homework, and expanded school-based after-school services. The policy targeted private, off-campus academic tutoring. It did not restrict tutoring organized by schools; in fact, it encouraged schools to provide more after-school academic support. China therefore offers an unusually clear case of a state-led attempt to substitute public provision for private tutoring.

This policy raises a question that is central to education and development: can public provision reduce inequality in access to supplementary education? The answer depends on which part of the tutoring market is measured. If school-based services expand in rural areas while private tutoring contracts in cities, a combined measure of "any tutoring" will show convergence even if the private market remains unequal. Yet most surveys and policy debates rely on exactly this kind of aggregate measure. The result can be a misleading picture of policy success: public provision expands, the aggregate gap narrows, but the market-based advantage that shadow education represents persists.

This paper separates the two margins. Using the child module of the China Family Panel Studies (CFPS) for 2018, 2020, and 2022, we distinguish school-based tutoring (`wt2`) from private/other tutoring (`wt7`). We document three facts. First, the aggregate tutoring gap narrowed: the rural–urban gap in any tutoring fell from 19.8 percentage points in 2018 to 6.5 points in 2022. Second, the convergence was almost entirely school-based: the school-based gap fell from 14.0 to 2.2 points, while the private tutoring gap fell only from 7.5 to 4.9 points. With family controls, the private gap was 3.9 points in 2018 and 2.2 points in 2022, and the change is statistically insignificant. Third, the spending gap remained large: rural households still spent far less on tutoring than urban households.

The paper makes three contributions to debates on education and development. First, it provides national panel evidence on the rural–urban gap in private tutoring, a dimension of educational inequality that is usually studied with local or cross-sectional data. Second, it shows that the aggregate gap narrowed because public school-based provision expanded, not because private tutoring converged. This is a measurement point with substantive consequences: evaluations that use "any tutoring" can attribute a public-sector expansion to the private market. Third, it draws out the policy implication that public provision can compress access gaps without equalizing the private market or the quality of provision. That distinction matters for developing countries considering restrictions on shadow education.

The paper is descriptive rather than causal. The double reduction policy was national, and the CFPS does not provide a clean control group or provider-level data. We do not attribute the 2022 change to the policy. The pattern is consistent with the policy period and with the expansion of school-based services, but pandemic-related disruption, measurement changes, and differential trends cannot be ruled out. The contribution is to establish a robust decomposition that any causal evaluation must confront.

The rest of the paper is organized as follows. Section 2 provides the policy context and conceptual framework. Section 3 describes the data and measures. Section 4 presents the analytical approach. Section 5 reports the findings. Section 6 presents robustness checks and

alternative explanations. Section 7 discusses the implications for educational development. Section 8 concludes.

1.1 Contribution to debates on shadow education and educational inequality

This paper connects three literatures. The first is the economics and sociology of shadow education. Bray and Kwok (2003) conceptualize private tutoring as a response to high-stakes examinations and family competition, and Dang and Rogers (2008) show that tutoring can deepen human capital inequality when access is unequal. Buchmann, Condron, and Roscigno (2010) show that shadow education is stratified by family resources even in a high-income context. Most of this literature is cross-sectional or single-country, and relatively little is known about how the rural–urban gap responds to a large supply-side shock.

The second is the literature on high-stakes testing and household investment. High-stakes exams can increase pressure on students and families (Högberg & Horn, 2022; Wang, 2016), and exam failure can have lasting psychological and educational costs (Beck et al., 2024; Machin et al., 2020). This literature has focused mainly on student outcomes and stress. It has paid less attention to the household investment margin: when the rules of the exam change, families may reallocate time and money toward tutoring, and the reallocation may differ by rural status. Our paper treats tutoring as a household investment that responds to the perceived returns to exam preparation.

The third is the literature on educational inequality in China. Hannum and Wang (2006) document persistent rural–urban gaps in educational access and attainment. Shadow education is a plausible mechanism behind these gaps, but national evidence is limited. Existing Chinese studies often use local surveys, focus on urban families, or describe the growth of the tutoring industry without a rural comparison. The double reduction policy provides an unusually sharp change in the supply of private tutoring, making the rural–urban margin especially informative. Recent evaluations document the policy's supply contraction and its effects on tutors and student well-being (Huang et al., 2025; Zhou & Fan, 2025), but they do not separate public school-based provision from private tutoring.

Our contribution is threefold. First, we provide national panel evidence on the rural–urban gap in private tutoring using consistent measures across 2018, 2020, and 2022. Second, we show that the aggregate gap narrowed because school-based provision expanded, not because private tutoring converged. Third, we distinguish participation, hours, and spending, and show that access to school-based support converged while the private spending gap persisted. The last point is important: public provision can equalize access without equalizing the private market.

2. Policy context and conceptual framework

2.1 Shadow education and the public–private divide

Shadow education is private, out-of-school supplementary tutoring (Bray, 2009). It includes commercial tutoring companies, small local providers, one-on-one tutoring, online courses, and informal tutoring arranged by families. It does not include tutoring organized by schools as part of the school day or as an after-school service.

The distinction is not semantic. Private tutoring is purchased in a market, priced by providers, and rationed by household income and access. School-based tutoring is provided by schools, often at low or zero marginal cost, and allocated by school policy. The two margins have different equity implications. If private tutoring expands, household inequality may widen. If school-based tutoring expands, public provision may compress inequality.

In China, both margins grew before 2021. Private tutoring became a large industry, especially in cities. School-based after-school services also expanded, particularly for primary and junior-high students. The double reduction policy targeted the private margin while explicitly encouraging the public margin. A combined measure of "any tutoring" therefore mixes two treatments with opposite policy intentions.

2.2 China's double reduction policy

In July 2021, the Central Committee of the Communist Party of China and the State Council issued the "Opinions on further reducing the burden of homework and off-campus training for students in compulsory education" (Central Committee of the Communist Party of China & State Council, 2021). The policy is commonly known as "double reduction" because it targets both homework and off-campus training.

The main provisions relevant to this paper are:

1. **Supply restrictions.** No new for-profit institutions providing academic tutoring to students in compulsory education; existing institutions must register as non-profit and cannot raise capital or be listed.
2. **Time restrictions.** Academic tutoring is prohibited on weekends and during holidays and vacations.
3. **School services.** Schools are encouraged to provide after-school services, including homework help and academic support.
4. **Demand-side signals.** The policy publicly discourages excessive tutoring and emphasizes school-based learning.

The policy was implemented nationally, but enforcement varied across provinces and localities. Urban areas, where the private tutoring industry was concentrated, experienced the largest supply contraction. Rural areas had thinner private markets to begin with, and school-based services may have expanded.

2.3 Conceptual framework

Consider a household that chooses whether and how much tutoring to purchase for a child. The household weighs the perceived return to tutoring against its price, income, and the opportunity cost of the child's time. The perceived return depends on the exam system, the child's baseline achievement, and the family's information about admissions. The price includes the direct fee, travel time, and the cost of finding a provider. Rural households face higher travel costs, thinner provider markets, and lower incomes, so they purchase less private tutoring at any given perceived return.

The double reduction policy can shift both the price and the choice set. In urban areas, the policy restricts the supply of for-profit academic tutoring. If enforcement is effective, the price of formal private tutoring rises and the quantity falls; some families may substitute toward informal or one-on-one tutoring, which is harder to regulate. In rural areas, the private market was thinner to begin with, so the direct supply restriction is smaller. At the same time, school-based after-school services may expand, lowering the effective price of academic support.

This framework yields three implications. First, if the policy binds mainly in urban private markets, urban private participation should fall more than rural private participation, and the private gap should narrow. Second, if school-based services expand in rural areas, rural school-based participation should rise, and the school-based gap should narrow. Third, if urban families substitute toward higher-quality or informal private tutoring, the private spending gap should persist even if participation converges.

The distinction between the two margins is therefore central. A finding that total tutoring converges is not evidence that shadow education converges. It may simply mean that public provision expanded while the private market remained unequal.

3. Data and methods

3.1 Data

The China Family Panel Studies (CFPS) is a nationally representative longitudinal survey conducted by the Institute of Social Science Survey at Peking University (Xie & Hu, 2014). The child module covers children aged 10–15 and includes detailed information on schooling, tutoring, family environment, and child outcomes.

We use the 2018, 2020, and 2022 waves. The 2016 child module asks a different set of tutoring questions and is not directly comparable with later waves, so we exclude it. The raw child sample contains 10,094 children across all three waves; the 2018–2022 sample contains 7,578. After requiring non-missing parental education and family income, the main estimation sample contains 7,145 children (94.3% of the 2018–2022 sample). Retention is similar across rural and urban groups. The analytic sample contains 28 province identifiers. Because the CFPS baseline design covers 25 provinces and later waves include resampling, we do not interpret province-level estimates as representative of all provinces.

The sample is an unbalanced repeated cross-section: children enter and leave the 10–15 age window across waves, and the CFPS is a panel rather than a school-based survey. We do not use sampling weights in the main specification because the estimand is the conditional rural–urban gap rather than a population mean. The child fixed-effects specification uses the subset of children observed in more than one wave.

3.2 Measures

Tutoring. The child module asks whether the child attended school tutoring (`wt2`) and whether the child attended other tutoring (`wt7`). School tutoring refers to tutoring organized by or at school. Other tutoring refers to private or external tutoring; this is our measure of shadow education. We also construct any tutoring as one if either is yes. Weekly hours are measured

separately for school tutoring (`wt201`) and other tutoring (`wt701`); we sum them for total academic hours. Spending is the reported cost of extracurricular/tutoring in the past 12 months (`wd503r`), converted to inverse hyperbolic sine. The `wt7` item asks about "other tutoring classes" and does not separately identify academic versus non-academic private tutoring; `wd503r` is a general extracurricular/tutoring spending item and may include non-academic classes. We therefore interpret private/other tutoring as reported private tutoring, and spending as extracurricular/tutoring-related spending, not as a pure price measure for academic shadow education.

Rural status. The CFPS provides a census-based urban/rural indicator. Rural is one minus urban. This is a residence-based measure, not a hukou-based measure. We discuss the implications below.

Family controls. Parental education is the maximum of father's and mother's years of schooling, constructed from the family roster. Family income is log per-capita net family income. We do not control for family expenditure in the main specification because tutoring spending is a component of total expenditure; including it would be a "bad control" that absorbs part of the outcome.

3.3 Descriptive patterns

Table 1 reports means by year and rural status. In 2018, 13.4% of rural children received any tutoring, compared with 33.2% of urban children—a gap of 19.8 percentage points. By 2022, the figures were 24.7% and 31.2%, a gap of 6.5 points.

The decomposition is more informative. The school-based gap fell from 14.0 points in 2018 (23.5% vs 9.5%) to 2.2 points in 2022 (23.1% vs 20.9%). The private tutoring gap fell only from 7.5 points (12.2% vs 4.7%) to 4.9 points (9.5% vs 4.6%). In other words, the aggregate convergence is driven by school-based provision, not by the private market.

Table 1. Tutoring by year and rural status

	2018 Urban	2018 Rural	2020 Urban	2020 Rural	2022 Urban	2022 Rural
Any tutoring	0.332	0.134	0.389	0.222	0.312	0.247
School-based	0.235	0.095	0.273	0.178	0.231	0.209
Private/ other	0.122	0.047	0.144	0.057	0.095	0.046
Academic hours/week	3.04	1.65	4.20	3.57	2.46	2.61
Tutoring spending (yuan/year)	2,653	312	3,257	432	1,957	494
N	1,184	1,623	1,029	1,341	978	1,156

Notes: CFPS child module, age 10–15. Any tutoring = school-based or private/other. School-based = tutoring organized by or at school. Private/other = private or external tutoring. Academic hours = school tutoring hours + other tutoring hours. Spending is the reported cost of extracurricular/tutoring in the past 12 months. The table reports unweighted means.

4. Analytical approach

We estimate an event-study gap decomposition separately for total, school-based, and private tutoring:

$Y(ipt) = a + b_2020\ (Rural_i \times Year_2020) + b_2022\ (Rural_i \times Year_2022) + X(ipt)'g +$ province FE + year FE + e(ipt).

Here Y(ipt) is the outcome for child i in province p in year t, Rural_i is an indicator for rural residence, X(ipt) includes age, gender, parental education, and log family income, and e(ipt) is an error term. The omitted interaction is 2018, so b_2020 and b_2022 capture changes in the rural-urban gap relative to 2018. Standard errors are clustered at the province level.

The coefficient of interest is b_2022. A positive coefficient means the rural gap narrowed (or reversed) relative to 2018. We report the implied gap in each year as the baseline rural coefficient plus the relevant interaction.

The specification is descriptive. It does not exploit a randomized or quasi-random assignment of the policy. The estimation variation is the differential change in rural and urban outcomes over time, conditional on province and year fixed effects and family controls. This design absorbs permanent differences between provinces and common shocks, but it cannot rule out province-specific trends or shocks that differentially affect rural and urban families.

We also estimate a child fixed-effects specification for private and school-based tutoring:

$Y(it) = a_i +$ year FE $+ b_2020\ (Rural_i \times Year_2020) + b_2022\ (Rural_i \times Year_2022) + X(it)'g + e(it),$

where a_i is a child fixed effect. This absorbs time-invariant child and family characteristics. Because rural status is time-invariant in the main specification, the main effect of rural is absorbed; the interaction terms are identified by within-child changes over time.

5. Findings

5.1 The aggregate gap and its composition

Table 2 reports raw and conditional gaps in Panel A and regression estimates in Panel B. Column (1) uses any tutoring; column (2) uses school-based tutoring; column (3) uses private/other tutoring. All models include province and year fixed effects, age, gender, parental education, and log family income.

Table 2. Raw and conditional rural–urban gaps, and regression estimates

	(1) Any tutoring	(2) School-based	(3) Private/other
A. Gaps (pp)			
Raw gap, 2018	19.8	14.0	7.5
Raw gap, 2022	6.5	2.2	4.9
Conditional gap, 2018	10.5	7.4	3.9
Conditional gap, 2022	−0.3	2.1	2.2
B. Estimates			
Rural (2018 baseline)	−0.105***	−0.074***	−0.039***
	(0.018)	(0.016)	(0.012)
Rural × 2020	0.022	0.040	−0.017
	(0.024)	(0.026)	(0.017)
Rural × 2022	0.102***	0.094***	0.016
	(0.024)	(0.022)	(0.012)
Province FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	7,145	7,056	7,139
R^2	0.121	0.097	0.046

Notes: Raw gaps are differences in means from Table 1. Conditional gaps are the baseline rural coefficient plus the relevant interaction from Panel B, conditional on province and year fixed effects, age, gender, parental education, and log per-capita family income. Standard errors clustered by province in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The raw total gap narrows from 19.8 points in 2018 to 6.5 points in 2022 (Table 1). Conditional on family background and province fixed effects, the gap narrows from 10.5 points to essentially zero in 2022 (−0.3 points). The school-based gap narrows from 7.4 points to a small positive gap (+2.1 points). The private gap narrows from 3.9 points to 2.2 points, but the 2022 interaction is small and statistically insignificant. The aggregate convergence is therefore concentrated in school-based provision. The private 2022 interaction is small (0.016, 95% CI −0.009 to 0.042), and the implied conditional gap is 2.2 percentage points. The confidence interval cannot reject zero, but it also cannot rule out changes of roughly ±4 percentage points. We therefore interpret the result as no evidence of private convergence, not as proof that the private gap was unchanged. The child fixed-effects estimate is larger (0.044, 95% CI −0.003 to 0.092, $p = 0.069$), which is consistent with a modest but imprecisely estimated narrowing.

Figure 1 shows the decomposition between school-based and private tutoring. The private lines are flat: urban private participation falls from 12.2% to 9.5%, and rural private participation stays around 4.6–4.7%. The school-based lines converge.

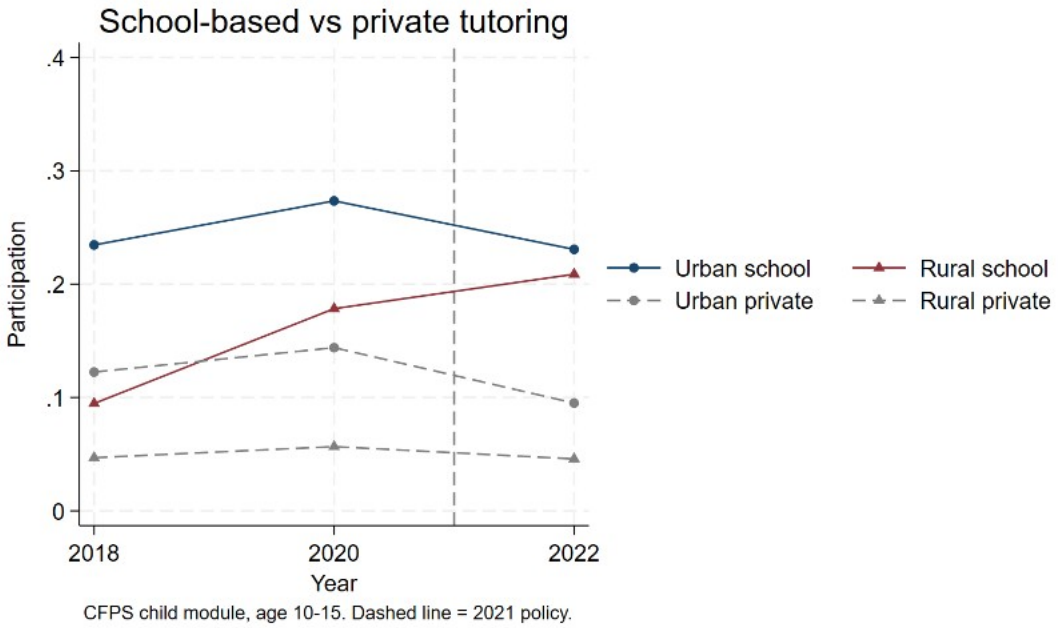


Figure 1. School-based and private tutoring by year and rural status, CFPS child module, age 10 –15. Dashed line marks 2021.

5.2 Child fixed effects

Table 3 reports fixed-effects estimates. The child fixed-effects sample is restricted to children observed in at least two waves (4,527 observations for private tutoring; 4,481 for school-based tutoring; 2,119 and 2,113 children, respectively). For private tutoring, the 2022 interaction is +0.044 (SE 0.024, p = 0.069); for school-based tutoring, it is +0.096 (SE 0.037, p = 0.009). The school-based convergence survives within-child comparisons; the private convergence remains small and only marginally significant. Because the fixed-effects sample requires at least two waves, it is skewed toward children who were 10–13 in 2018; the close correspondence between the fixed-effects estimate (0.096) and the pooled estimate (0.094) is reassuring, but the two estimates are not from the same sample. This reinforces the decomposition: the robust convergence is in public provision, not in the private market.

Table 3. Fixed effects, hours, and spending

	(1) Private/other	(2) School-based	(3) asinh(hours)	(4) asinh(spending)
A. Child FE				
Rural × 2020	0.003	0.024		
	(0.020)	(0.029)		
Rural × 2022	0.044*	0.096***		
	(0.024)	(0.037)		
Child FE	Yes	Yes		
N	4,527	4,481		
B. OLS				
Rural (2018 baseline)			−0.230***	−1.023***
			(0.050)	(0.160)
Rural × 2020			0.069	−0.015
			(0.089)	(0.183)
Rural × 2022			0.265***	0.713***
			(0.067)	(0.188)
N			7,111	7,120

Notes: Columns (1)–(2) report child fixed-effects estimates restricted to children observed in at least two waves, with year fixed effects and log family income; time-invariant characteristics are absorbed by the child fixed effects. Standard errors are clustered by child. Columns (3)–(4) report OLS estimates with province and year fixed effects, age, gender, parental education, and log income; standard errors are clustered by province. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Hours and spending

Total academic hours converge: the 2018 gap is −0.230 in asinh units, and the 2022 interaction is +0.265, implying a small positive gap (+0.035). This is consistent with the school-based expansion.

Spending tells a different story. The 2018 rural gap is −1.023 in asinh units, and the 2022 interaction is +0.713, leaving a gap of about −0.310. The raw means in Table 1 show a much larger difference—494 versus 1,957 yuan in 2022, or about 75% lower—because the asinh model compresses zeros and evaluates the gap at the mean of controls. Both measures point to a large spending gap. Rural households caught up in participation and hours but still spent substantially less on tutoring. This is consistent with a shift toward cheaper, school-based provision in rural areas, while urban families continued to purchase more expensive private services.

6. Robustness and alternative explanations

6.1 Excluding 2020

The 2020 wave coincides with the COVID-19 pandemic, and excluding it does not solve the pandemic confound: it leaves a comparison between 2018 and 2022 that straddles the pandemic. We use the exercise only to check whether the 2022 result is driven by the anomalous 2020 level. For private tutoring, the 2022 interaction remains small and insignificant; for school-based tutoring, it remains positive and significant. The decomposition is therefore not mechanically driven by the 2020 wave, but the pandemic remains an unresolved alternative explanation.

6.2 Timing of the school-based convergence

The timing of the school-based convergence is more nuanced than a simple 2021 break. The raw school-based gap was 14.0 points in 2018, 9.5 points in 2020, and 2.2 points in 2022. About 38% of the total decline (4.5 of 11.8 points) had already occurred by 2020, before the double reduction policy. Rural school-based participation rose from 9.5% in 2018 to 17.8% in 2020 and 20.9% in 2022; about 73% of the rural increase (8.3 of 11.4 points) occurred before 2020. Between 2020 and 2022, the school-based gap fell by 7.3 points: 4.2 points from the decline in urban school-based participation and 3.1 points from the rise in rural participation.

This pattern is more consistent with a continuation of rural after-school expansion that began before 2021 than with a single policy break. The 2021 policy may have contributed to the 2022 decline in urban school-based participation, but it cannot explain the earlier rural expansion. We therefore interpret the school-based convergence as a public-provision phenomenon with a longer timing, not as a clean double-reduction effect.

6.3 Family background

Family background is strongly associated with tutoring. Each additional year of parental education is associated with a higher probability of any tutoring, and higher per-capita family income is associated with more tutoring. These gradients are consistent with tutoring as a household investment. Adding family controls to a province-fixed-effects model reduces the conditional 2018 total gap from 15.8 to 10.5 percentage points, but does not explain the 2022 convergence. The private gap is smaller than the total gap in both years, and its change remains insignificant.

6.4 Measurement and inference

The 2018–2022 child module uses consistent item stems for school and other tutoring. The 2016 module differs and is excluded. Tutoring is reported by the child or a proxy parent; proxy reports may understate private tutoring, especially after the policy increased social desirability. If underreporting is stronger in urban areas after 2021, the private gap would be overstated, not understated. This makes the null result for private convergence conservative.

The standard errors are clustered at the province level with 28 clusters. Inference should be interpreted with caution; wild cluster bootstrap would be preferable with so few clusters. We report conventional cluster-robust standard errors and treat marginal significance ($p < 0.10$) as suggestive only.

6.5 Non-academic placebo

If the school-based convergence reflects academic support rather than a general expansion of all extracurricular activities, non-academic classes should not show the same pattern. The child module asks whether the child attended hobby classes (`wt4`), which are exempt from the double reduction restrictions. The estimates use the same specification and sample as Table 2, with province and year fixed effects, age, gender, parental education, and log family income (N = 7,141). In 2018, rural children were 4.7 percentage points less likely to attend hobby classes than urban children (SE 1.9). The 2022 interaction is −2.5 percentage points (SE 2.4, $p = 0.307$), so the non-academic gap did not converge. This placebo supports the interpretation that the school-

based convergence is specific to academic tutoring, not a general increase in rural extracurricular participation.

The placebo is conservative. After 2021, many academic tutoring providers shifted toward non-academic classes, which could have raised urban non-academic participation and narrowed the hobby-class gap even without a change in rural access. The fact that the gap did not narrow makes the academic-specific interpretation more credible, not less.

6.6 Same-child transitions

The participation series are consistent with a public-substitute channel, but they do not directly show that the same children moved from private to school-based tutoring. We therefore examine children observed in both 2018 and 2022 (N = 583). Of the 47 children who received private tutoring in 2018 but not in 2022, 16 (34%) received school-based tutoring in 2022. The within-child correlation between the change in school-based participation and the change in private participation is −0.13. This is consistent with substitution at the child level, but the sample is selected—children must remain in the 10–15 age window—and the correlation is descriptive, not causal.

7. Discussion

7.1 Public provision as a substitute

The decomposition points to school-based provision, not private tutoring, as the source of the aggregate convergence. This is consistent with a public substitute: as schools expanded after-school academic support, families—especially rural families—could rely less on private tutoring. The double reduction policy restricted private, off-campus academic tutoring but encouraged schools to provide after-school services, so the policy environment reinforced this substitution.

The timing, however, suggests two phases rather than a single 2021 break. Rural school-based participation rose sharply between 2018 and 2020, before the double reduction policy, consistent with the earlier expansion of after-school services. Between 2020 and 2022, the school-based gap continued to close, but a larger share of that later decline came from urban school-based participation falling from its 2020 level. The aggregate convergence is therefore a public-provision phenomenon, but it is not solely a 2021 policy effect.

This distinction matters for interpretation. The results do not show that the double reduction policy equalized private tutoring. They show that public provision can compress the aggregate gap while the private market remains unequal. That is a more limited—and more defensible—claim.

7.2 The private gap

The private gap did not close. In 2018, urban children were 7.5 percentage points more likely to receive private tutoring than rural children; in 2022, the gap was 4.9 points. With family controls, the gap fell from 3.9 to 2.2 points, but the change is statistically insignificant. The private market remains unequal.

This matters for the interpretation of the double reduction policy. If the goal was to reduce the private tutoring advantage of urban families, the policy period did not eliminate it. Urban private

participation fell, but rural private participation did not rise; the gap narrowed mainly because urban participation fell, not because rural families gained access. The private market may have contracted, but it did not equalize.

7.3 Quantity versus quality

The coexistence of convergence in total participation and hours with a persistent spending gap is important. Access to tutoring has become more equal, but the quality and intensity of private tutoring may not have. Urban families may shift toward more expensive one-on-one or hidden tutoring, while rural families rely on cheaper school-based services. If spending is a proxy for quality, the policy may have equalized access without equalizing the returns to tutoring.

This distinction matters for policy. If the goal is to reduce the financial and time burden of tutoring, the policy appears to have succeeded in reducing urban private participation. If the goal is to equalize educational opportunity, the spending gap suggests that further supply-side investment in rural schools is needed.

7.4 Alternative explanations

Three alternative explanations deserve attention. The first is the pandemic. The 2020 wave was collected during a period of school disruption, and the 2022 wave followed reopening. Urban private tutoring markets may have been more disrupted than rural markets. The fact that the private gap remains in 2022, and that the school-based convergence survives excluding 2020, makes a pure pandemic story less likely, but it cannot be excluded.

The second is measurement. The child module relies on child or proxy reports. If urban parents became more reluctant to report private tutoring after the policy, the urban decline would be overstated. This is plausible because the policy stigmatized off-campus academic training. The spending data partially address this concern: if urban families had simply stopped reporting tutoring, spending would fall as well. Spending did fall, but the spending gap remained large, which suggests that the participation decline is not purely a reporting artifact.

The third is demographic and economic change. Rural incomes and parental education rose between 2018 and 2022, and these changes could increase rural demand for tutoring. Our family controls absorb part of this change, but not all of it. The convergence in school-based provision could therefore reflect rural development rather than policy. This interpretation is not mutually exclusive with the policy story: rising rural incomes may have enabled rural families to take advantage of school-based services that the policy expanded.

7.5 Implications for educational development

The findings have three implications for education policy in developing countries. First, aggregate measures of “any tutoring” are a poor guide to policies that regulate private tutoring. A government can restrict private providers and expand public after-school services, producing convergence in the aggregate while the private market remains unequal. Evaluations should measure public and private provision separately, and should report spending as well as participation.

Second, public provision can compress access without equalizing quality. The school-based gap closed, but the spending gap remained. If spending is correlated with instructional quality, school-based services may have reduced the quantity of private tutoring without equalizing the quality of academic support. This is a central caveat for governments that hope to reduce inequality through public after-school programmes.

Third, regulation alone is unlikely to be sufficient. The Chinese case suggests that supply restrictions can change the private market, but the persistence of the private spending gap implies that demand for advantage does not disappear. Policies that combine regulation with sustained investment in rural schools, teacher supply, and school-based academic support are more likely to reduce inequality than regulation alone. China's implementation capacity is unusually high; in lower-capacity settings, the public substitute may be harder to deliver, and private tutoring may move further into informal or hidden markets.

7.6 Limitations

This paper has several limitations. First, it is descriptive. The policy was national, and there is no clean control group. The estimates should not be interpreted as causal effects. Second, the 2016 wave is not comparable, so we have only two pre-policy waves and one post-policy wave. Third, the data are child-proxy reports, and private tutoring may be underreported. Fourth, family controls may be endogenous. Fifth, the standard errors are clustered at the province level, with only 28 clusters; inference should be interpreted with caution. Sixth, rural status is residence-based, not hukou-based; children who move between rural and urban areas may be misclassified. Seventh, the CFPS contains cognitive tests, but they are not comparable across all three waves: 2018 has word and math tests, 2020 has word recall and number series, and 2022 has a math test with substantial missingness. We therefore do not construct a consistent achievement outcome. This is a limitation relative to evaluations that can measure test-score convergence. Finally, we lack provider-level data on prices, quality, and informal tutoring.

8. Conclusion

This paper shows that the rural–urban gap in total tutoring in China narrowed sharply between 2018 and 2022, but the convergence was the wrong kind for understanding shadow education. The school-based gap fell from 14.0 to 2.2 percentage points, while the private tutoring gap fell only from 7.5 to 4.9 points and is statistically insignificant after family controls. The aggregate convergence reflects the expansion of public school-based provision, not convergence in the private market.

For education and development, the main lesson is that public provision can narrow access gaps without reducing market-based inequality. A combined measure of "any tutoring" can show convergence even when the private market remains unequal. Governments that regulate shadow education should separate school-based and private provision, and should track spending and quality, not only participation.

The Chinese case also cautions against treating regulation as a substitute for public investment. Investments in rural schools, teacher supply, and school-based academic support are likely to be necessary complements to supply restrictions. Future research should combine household data

with provider-level and school-level data, and should examine whether public after-school services reach the poorest children and improve learning, not only participation.

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Conflict of interest: The author declares no conflict of interest.

Data availability: The CFPS data are publicly available from the Institute of Social Science Survey, Peking University. Replication code and derived datasets are available from the author upon reasonable request.

Ethics: Not applicable. The study uses publicly available secondary data; no human participants were involved.

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Appendix: Variable definitions

wt2 asks whether the child attended school tutoring; wt7 asks whether the child attended other tutoring classes; wt201 and wt701 record weekly hours of school and other tutoring; wd503r records the cost of extracurricular/tutoring in the past 12 months. The private/other category is the closest available measure of out-of-school private tutoring; it does not separately identify academic versus non-academic providers.

Variable	Definition
Any tutoring	1 if child attended school tutoring (`wt2`) or private/other tutoring (`wt7`)
School-based tutoring	1 if child attended school tutoring (`wt2`)
Private/other tutoring	1 if child attended private or other tutoring (`wt7`)
Academic hours	Weekly hours of school tutoring (`wt201`) + other tutoring (`wt701`)
Spending	Reported cost of extracurricular/tutoring in past 12 months (`wd503r`), asinh
Rural	1 if census-based rural residence
Parental education	Maximum of father's and mother's years of schooling
Family income	Log per-capita net family income